

Sleep Duration as a Predictor of Depressive Symptoms Among University Students: A Cross-Sectional Quantitative Study

Abstract : This study adopts a quantitative cross-sectional design to examine whether average sleep duration predicts depressive symptom severity among university students. Data will be collected from approximately 100–120 participants using a structured online questionnaire. Sleep duration will be measured through self-report, while depressive symptoms will be assessed using the Patient Health Questionnaire (PHQ-9) (Kroenke et al., 2001).Data will be analysed using descriptive statistics, Pearson correlation, and linear regression ($p < .05$).

Introduction

Sleep is a critical neurobiological process supporting emotional regulation, cognitive performance, and neural restoration (Walker, 2017). Disruptions to circadian rhythms impair prefrontal–amygdala functioning, increasing vulnerability to negative emotional states. University students are especially prone to sleep deprivation due to academic pressure, social commitments, and digital exposure (Lund et al., 2010). Evidence suggests that individuals sleeping less than recommended levels (7–9 hours) demonstrate poorer emotional regulation and higher depressive symptoms (Hirshkowitz et al., 2015).

Research Aim

To investigate the relationship between average sleep duration and depressive symptoms among university students.

Hypotheses

- H1:** Sleep duration will be negatively correlated with depressive symptoms
- H2:** Sleep duration will significantly predict depressive symptom severity

Ethical Considerations

- Informed Consent:** Participants will receive full study details before participation
- Voluntary Participation:** Right to withdraw at any time without penalty
- Anonymity:** No identifiable data collected
- Confidentiality:** Data stored securely on password-protected systems
- Psychological Risk Management:**
 - PHQ-9 may induce discomfort
 - Participants will be provided with **mental health support contacts**
- Inclusion Criteria:** Age 18+
- Ethical Approval:** Institutional ethics committee approval required

References

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Research Design & Method

Design

- Quantitative
- Cross-sectional
- Correlational

Participants (N) = 100–120 university students

Sampling: Snowball sampling (non-prob.)

Measures

Sleep Duration: Self-reported average hours per night

•Depression:

- **PHQ-9 Scale**
- 9 items, 0–27 score range
- Measures severity of depressive symptoms (Kroenke et al., 2001)

Procedure

- Online survey (Google Forms/Qualtrics)
- Duration: 5–7 minutes
- Recruitment via social networks
- Data collection period: 2–3 weeks

Data Analysis

- Descriptive statistics (Mean, SD)
- Normality testing (Skewness/Kurtosis)
- Pearson Correlation (Sleep vs Depression)
- Linear Regression (Predictive relationship)
- Significance level: $p < .05$

Limitations & Critical Evaluation

- Self-report bias:** Sleep duration may be inaccurately estimated
- Cross-sectional design:** Limits causal inference
- Sampling bias:** Snowball sampling reduces generalisability